

Cross-Project Bug Localization: Consolidated Vetting List

To A Eashaan Rao. From Sridhar Chimalakonda. RISHA Lab, IIT Tirupati. Reference: the current 24-page manuscript. Target venue: Journal of Systems and Software.

The numbers are clean. Table 8 and Table 9 reconcile to Table 6 for all three models, every MRR sits under the ceiling its own Top-K values permit, the retrieval ceiling is respected, and no values from earlier versions survive anywhere. The BLAZE correction propagated correctly. What follows is everything still open, in four groups.

Part A must be done before submission. Part B changes how the paper reads to someone who has published in this area. Part C is what a technical referee will ask. Part D protects the empirical-standards claim the paper makes about itself. Change nothing that is not on this list.

Part A. Before submission

Where	What is wrong	What to change
Section 3.6, WP-small bullet, and Table 5	Both say WP-small is trained on ~20% of the target corpus, which restores the budget-matching claim the orchestration script disproved. The abstract says ~10% and Case 1 says 10%, so the paper contradicts itself, and the methods section states something the code contradicts.	Section 3.6 bullet, replace with: “WP-small (WPS) (Within-Project, Limited): trained on ~10% of the target corpus (12.5% of the same 80% pool that WP-large draws from, which neither the test set nor WP-large's remainder touches). This is the primary baseline throughout, a deliberate departure from prior work's ample-data comparator: measuring CPL's benefit against a data-scarce baseline tests whether it delivers on its design intent. CP-transfer draws 25% of that pool (~20% of the corpus), so the two conditions are not matched on target labels; Section 5.4 discusses the resulting confound.” Table 5, WP-small row, Training Data: “~10% target bugs only”.
Section 5.4, internal validity	The budget-mismatch paragraph has been deleted, so the strongest objection to RQ1 is now undisclosed.	Add after the budget-matching discussion: “CP-transfer and WP-small are not matched on target-project labels. WP-small draws 12.5% of the 80% target train-pool (≈ 10% of the corpus) while CP-transfer draws 25% (= 20%), twice WP-small's share, for example 11 against 22 bugs for pydata/xarray and 133 against 267 for qiskit/qiskit. A CP-transfer advantage therefore reflects both cross-project knowledge and twice the target supervision, and the two are not separable here. Disambiguating them needs a WP-small re-run at ~20%, which we did not carry out, so the CP-transfer-against-WP-small gains reported throughout should be read as an upper bound on the cross-project-knowledge component.” Then one sentence naming the fix and its size: “Restoring budget-matching requires a within-project baseline at ~20% of target labels, which is 39 target-only

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		runs (13 targets × 3 models) and no source-corpus training; we scope it as the immediate next step.”
Contribution C3	C3 advertises 1.6 / 15.9 / 31.7 percent as “directly actionable”, and Section 4.2.2's own robustness check then reports 1.6 / 6.3 / 20.6 against median WP-small baselines, and 0 / 1.6 / 0 under the most conservative band. A headline contribution that collapses under the paper's own sensitivity analysis is the easiest objection in the paper.	Replace with: “(C3) A negative-transfer risk spectrum across architectures under identical conditions: 1.6% / 15.9% / 31.7% of pairs for BLAZE / TRANP-CNN / COOBA against single WP-small runs, falling to 1.6% / 6.3% / 20.6% against per-target median baselines (Section 4.2.2), the first such rates reported for CPL bug localization.”
Section 3.3, Domain Classification Protocol	The new two-phase protocol claims a student volunteer as second annotator and 82% inter-rater agreement, and the corresponding threat has been removed from Section 5.4. Earlier versions stated there was no second rater.	Confirm this pass actually took place and that both annotators' per-project labels are in the replication package. If it did, write “an initial raw inter-rater agreement of 82% (Cohen's $\kappa = X$)” and add the labels to Zenodo. If it did not, the protocol description comes out and the earlier threat goes back in.
Section 3.3, same paragraph	“moderated by the another co-author”.	“moderated by a third co-author”
Reference list, Ralph entry	Reads “Ralph, P., et al., 2021 ... aCM SIGSOFT Empirical Standards initiative.” Lowercase a, and Elsevier style does not use “et al.” in the reference list.	“Ralph, P., Baltes, S., Adisaputri, G., et al., 2021. Empirical standards for software engineering research. arXiv:2010.03525.”

Part B. Overclaims and framing

Where	What is wrong	What to change
Section 5.2, chronological cross-check, and Section 3.2	TRANP-CNN is now cited as Huo et al. 2021 throughout, but Section 3.2 lists it as 2019 and Section 5.2 argues “COOBA (2020) ... than TRANP-CNN (2019)”. With the corrected year the argument reverses: TRANP-CNN becomes the newer model, so its better outcome is exactly what publication year would predict. The cross-check refutes itself.	In Section 5.2: “A cross-check on timing: COOBA (IJCAI 2020) employs a more sophisticated adversarial mechanism than TRANP-CNN (first available 2019, TSE issue 2021), yet achieves worse outcomes across nearly all metrics.” Use the same parenthetical in Section 3.2's co-variation list. Alternatively drop the cross-check; the other three arguments in 5.2 carry the point without it.
Section 3.6, opening	“a ceiling data-scarce practitioners cannot reach, so beating it tells them nothing useful.” Dismisses every prior CPL paper in one clause, and one of the reviewers may have written one of them.	“a ceiling data-scarce practitioners cannot reach, so beating it leaves their question unanswered.”

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Section 1, Ch2	“any comparison against it answers the wrong question.”	“any comparison against it answers a different question from the one practitioners face.”
Section 1, third paragraph	“... establish that CPL surpasses a standard they cannot reach, never whether it beats the limited training they can perform.” The absolute “never” invites a reviewer to produce one counterexample and discredit the framing.	“... show that CPL surpasses a standard they cannot reach, not whether it beats the limited training they can perform.”
Section 2.2	Each prior paper gets a “but” clause in sequence: TRANSP-CNN “but evaluated only against ...”, BL-GAN “It does not test whether ...”, BLAZE “but tests only a single architecture”. Three in a row reads as score-settling rather than gap analysis.	Keep the TRANSP-CNN and BLAZE clauses, drop the BL-GAN one, and let the consolidated gap statement that follows carry the argument.
Section 4.2.3, COOBA paragraph	“contradicting the naive expectation that adversarial adaptation helps most within-domain”. Calling COOBA's design rationale naive is a direct dig at its authors.	“contrary to the expectation that adversarial adaptation would help most within-domain”
Section 4.3.2, BLAZE	“trivially zero predictive power, so practitioners can select sources freely without within-project evaluation.” “Trivially” is a needless flourish on $p = -0.00$, and the recommendation is stated without noting it rests on one model in one corpus.	“$p = -0.00$ ($p = 0.999$): no predictive power in this corpus, so source selection need not be preceded by within-project evaluation for BLAZE.”
Section 2.2, closing	“The conditions governing CPL success identified here, domain alignment, embedding quality, inference-time adaptation”. Embedding quality was never measured as a moderator, the domain-gap metric was non-significant for all three models, and BLAZE's domain effect is now 2.1 points. The sentence promises three identified conditions and the paper delivers one and a half.	“The conditions associated with CPL success here, chiefly target label volume and inference-time adaptation mechanism, may nonetheless inform ...”
Abstract and Section 1, opening sentences	“critical precursor” and “critical, labor-intensive activity” in consecutive openings.	Keep “critical precursor” in the abstract; Section 1 becomes “a labor-intensive activity in software maintenance”.

Part C. Technical review

Where	What is wrong	What to change
Table 2, Section 3.6	Early stopping “monitors validation loss”, but the scenarios define only a training share and a 20% test set. No validation split is defined anywhere. If validation comes out of the training share, WP-small's effective training set is smaller than stated (11 bugs for xarray, minus validation). If it comes from the test set, that is leakage and every number is affected. A referee will ask this first.	State it in Section 3.6: “A fixed 10% of each scenario's training data is held out for validation and early stopping; the 20% test split is never used for model selection.” Adjust the stated WP-small counts if validation is carved from them.

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Section 4.2.2, negative-transfer definition	Negative transfer is defined at a ± 0.01 MRR band while the measured run-to-run spread is 0.05 to 0.07 MRR. The threshold sits five to seven times inside the noise, so the primary definition classifies noise as negative transfer by construction.	Either define the band from the noise floor and report the ± 0.01 version as the permissive variant, or add: “The ± 0.01 band is a sensitivity threshold, not a noise-calibrated one; rates under noise-calibrated bands are reported above.”
Section 3.2 and Table 2, hyperparameters	No hyperparameter search for any model, and COOBA's adversarial weight is fixed at $\lambda_{\text{adv}} = 0.1$, a parameter known to be sensitive. COOBA's near-zero cold-start is the paper's most quotable negative result, so a referee will say it may be a tuning artifact.	Add to Section 5.4's re-implementation paragraph: “No hyperparameter search was run for any model; each uses the values reported in its original paper, including COOBA's $\lambda_{\text{adv}} = 0.1$. COOBA's profile may therefore partly reflect an untuned adversarial weight rather than the paradigm, a sensitivity we did not test.”
Section 3.7, MAP definition	The definition assumes every ground-truth file has a rank. For a bug with three ground-truth files where only one survives retrieval and reranking, it is undefined whether k is 3 with the missing terms contributing 0, or 1. The two give materially different MAP.	Add the clause: “... where k is the number of ground-truth files for the report and files absent from the reranked list contribute 0 to the sum.”
Section 3.7, Top-K Recall	Still reads “fraction of bug reports for which the ground-truth file appears within the top K”, singular, and the next sentence says MRR and MAP use multi-relevant definitions, which draws attention to Top-K not doing so.	“fraction of bug reports for which at least one ground-truth file appears within the top K positions”
Table 11, Section 4.3.3	Hit@1 and Kendall τ are computed over 13 targets. Every Hit@1 value is a multiple of 1/13, so 15.4% against 46.2% is two targets against six. No confidence intervals, no test, yet per-model conclusions are drawn.	Add to the caption: “n = 13 targets with at least three candidate sources; Hit@1 differences of one or two targets are not distinguishable at this sample size.”
Table 11, % oracle column	“74 to 93 percent of oracle with any heuristic” has no comparator. Without the expected recovery under random source selection the number cannot be read as evidence that selection is low-stakes. It is computable from data already in hand.	Add a random-selection row per model: mean CP-transfer MRR across the available sources divided by the best, averaged over targets. If random recovery is also near 80 percent, say so; the conclusion survives and becomes defensible.
Section 3.9, Cohen's d	The paper notes it uses “the same conventional thresholds used for the independent-samples d” but never gives the formula. For paired data, d_z (mean difference over SD of differences) is routinely two to three times larger than the independent-samples d, so judging d_z against 0.2 / 0.5 / 0.8 overstates the effect. BLAZE's $d = 1.16$ with a 95.2% win rate is consistent with d_z .	State the formula and acknowledge it: “Cohen's d here is d_z, the mean paired difference divided by the SD of the differences; because d_z exceeds the independent-samples d for the same data, the conventional thresholds should be read as a comparability convention rather than a strict benchmark.”

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Section 4.2.2 and Figure 5	Roughly 24 Spearman tests are run (eight features by three models) with no correction, while RQ1 applies Bonferroni within five-metric families. The inconsistency is visible. $\rho = -0.77$ survives anything; the LoC-ratio result at $p = 0.35$ would not.	Add: “These correlations are exploratory and uncorrected across the eight features and three models; only the target-label-volume associations survive a Bonferroni adjustment at this family size.”
Section 5.4, conclusion validity	Pairs are non-independent ($ICC \approx 0.31$ across 13 target clusters), but the primary RQ1 tests are Wilcoxon on all 63 pairs and the mixed-effects model appears only as a secondary check.	Keep the Wilcoxon as primary but state the choice: “The Wilcoxon tests treat pairs as independent; the mixed-effects model above is the check on that assumption, and it preserves the model-identity effect.”
Section 3.5, FAISS baseline	The first-stage FAISS MRR of 0.029 is far below what dense retrieval typically reports on this task, and it bounds every downstream number. The chunk-averaging explanation is plausible but untested, and with no BM25 run it cannot be sanity-checked.	Add after the 0.029: “This is low relative to reported first-stage performance in the IR-based localization literature; we attribute it to chunk-embedding averaging over long files, but without a lexical comparator the retrieval stage's quality is not independently verified (Section 5.4).”
Section 4.2.4, commutativity	The 16/28, 16/28 and 17/28 consistency rates are correctly described as weak, but no test is reported.	Add the one-sided binomial p-values in a parenthesis: 16/28 gives $p = 0.29$ and 17/28 gives $p = 0.17$. This makes the existing honesty explicit at the cost of one clause.

Part D. ACM SIGSOFT Empirical Standards

Where	What is wrong	What to change
Replication package	Section 3 and the Data Availability statement both assert that a completed checklist against the General and Data Science Standards is in the Zenodo package. If it is absent or a stub, that is a false statement in a submitted paper.	Confirm the completed checklist is in the deposit and that the DOI resolves to the version containing it, before submitting.
The checklist itself	The study does not satisfy several Data Science attributes: no preregistration, a post hoc practical-benefit criterion, random rather than chronological splits, a baseline that is not budget-matched, and a single seed. All are disclosed in the paper's prose, so a checklist claiming full compliance would contradict the paper.	Mark each as not met, with a one-line pointer to the section where it is discussed. Partial compliance honestly reported scores better than claimed completeness.
Data split (same as the technical item above)	Both standards ask for the training, validation and test partitions. The validation split is not described, which is a direct essential-attribute miss rather than a judgment call.	Fixed by the Section 3.6 addition above; once that is in, the attribute is met.

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Acknowledgements	Section 3.3 now says a student volunteer independently classified all 13 projects. The volunteer is not named anywhere, and the Acknowledgements thank only the institute.	“We thank [name] for independently classifying the project domains reported in Section 3.3.” Consider authorship if the contribution went beyond labeling. No ethics approval is needed for a co-rater, but if institute rules treat volunteer raters as participants, say so in one line.
Choice of standard	The data collection is repository mining over SWE-bench and BeetleBox, and the Empirical Standards collection carries a separate checklist for that alongside Data Science. Since the paper names its standards explicitly, the claim is checkable.	Confirm which checklist covers the mining component and either add it or satisfy yourself that Data Science covers it. If it applies, its sampling-frame and data-quality attributes are already met in substance by Section 3.3 and the label-noise threat, so this is a naming fix.

Sridhar Chimalakonda

The original ideas, analysis, and research synthesis in this report are the intellectual contribution of the authors. AI-assisted tools were used to enhance the clarity and presentation of the writing.